Supplementary Material

Particle Creation Bounds for Inhomogeneous Klein–Gordon Fields

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Numbered equations in this supplement are labeled (S1)–(S8). The repository README (README.md) gives the Python environment and reproduction commands.

S1. Bogoliubov–Valatin equations

For the oscillator $\hat{H}(t) = p^2/2 + \Omega^2(t)x^2/2$, the instantaneous Bogoliubov coefficients satisfy Eq. (11) with $u(0) = 1$ and $v(0) = 0$. We independently integrate the classical mode equation

$$\ddot{f} + \Omega^2(t)f = 0, \qquad f(0) = \frac{1}{\sqrt{2\Omega_0}}, \qquad \dot{f}(0) = -i\Omega_0 f(0). \tag{S1}$$

The instantaneous occupation is

$$N_{\mathrm{ex}}(t) = \frac{|\dot{f}|^2 + \Omega^2(t)|f|^2}{2\Omega(t)} - \frac{1}{2}. \tag{S2}$$

The integrations agree to 10^{-8}, the accuracy set by output interpolation. The Wronskian $\mathrm{Im}(f^*\dot{f})$ and the invariant $|u|^2 - |v|^2$ are conserved to 6×10^{-11} and 2×10^{-11}, respectively. The solver is SciPy's RK45 with relative tolerance 10^{-11} and absolute tolerance 10^{-13}.

S2. Single-mode tests for Figure 2

The Figure 2 calculation (code/fig2.py) starts from the instantaneous vacuum and uses the mode equation and solver settings in S1.

- **Panel (a).** $\Omega(t) = 1 + 0.02\sin(\omega_m t)$ on $0 \le t \le 100\pi$. At $\omega_m = 2$, $N_{\mathrm{ex}}(100\pi) = 133.25$, compared with the Mathieu estimate $\sinh^2(0.01 \cdot 100\pi) = 133.37$.
- **Panel (b).** A tanh ramp from 1 to 0.5 has centre $t_0 = 3$ and width $\tau = 0.3$. The frequency is then held constant and the evolution is integrated to $t = 20$.
- **Panel (c).** Tanh ramps from 1 to $r \in \{0.5, 0.25, 2\}$ use $\tau \in \{0.05, 0.1, 0.2, 0.3, 0.5, 0.8, 1.2, 2\}$. Corollary 1 also uses $\tau = 5$. The centre is $t_0 = 6\tau + 2$ and the final time is $2t_0 + 5$. The sudden-quench reference is $(1 - r)^2/(4r)$.

S3. Multimode and field tests

Mode matrix and coupled protocols

Write $A = V^{1/2}$ and $\hat{x}(t) = F(t)\hat{a}_0 + \overline{F}(t)\hat{a}_0^\dagger$. The initial conditions are $F(0) = A(0)^{-1/2}/\sqrt{2}$ and $\dot{F}(0) = -iA(0)F(0)$. The multimode tests use RK45 with relative tolerance 10^{-10} and absolute tolerance 10^{-12}. The instantaneous Bogoliubov matrix and total occupation are

$$\beta = \frac{A^{1/2}\overline{F} + iA^{-1/2}\dot{\overline{F}}}{\sqrt{2}}, \qquad N_{\mathrm{tot}} = \| \beta \|_F^2. \tag{S3}$$

The squeezing parameters are $r_k = \mathrm{arsinh}\sigma_k(\beta)$ and determine d_G through Theorem 1. We compute B from central differences of A with step 10^{-5} and use the trapezoidal rule for Z, L_{FS}, and Θ on about 3000 time points.

For the chain (code/chainbig.py), $V = c(t)I + K$, where K is tridiagonal with diagonal 0.4 and off-diagonal -0.2. We use $c(t) = [1 + 0.25\left(1 + \tanh\big((t-3)/0.3\big)\right)]^2$ and integrate the fixed normal modes separately to $t = 8$.

The four values reported for each chain are the excitation, the sum of Eq. (14) over modes, the square of the right side of Eq. (17), and the bound in Eq. (16), in that order.

- 50 modes give 0.82, 1.377, 1.384, and 2.1.
- 200 modes give 3.29, 5.51, 5.54, and 26.
- 1000 modes give 16.5, 27.6, 27.7, and 8.7×10^3.

The two-mode rotations in Appendix C.2 use

$$V(t) = R(\phi)\mathrm{diag}(1,9)R(\phi)^{\mathsf{T}}, \quad R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}, \quad \phi(t) = \kappa\tanh(t-3) \tag{S4}$$

with $\kappa = 0.5, 2, 10$ and final time 8. Other protocols use diagonal entries $V_{11} = 1$ and $V_{22} = 2.25$ except when V_{11} is modulated. They use $V_{11}(t) = 1 + 0.05\sin(2t)$ with $J = 0.3$ to $t = 60$, the coupling ramp $J(t) = 0.45[1 + \tanh\big((t-3)/0.05\big)]$ to $t = 8$, or $J(t) = 0.08\cos(2.5t)$ to $t = 80$. The off-resonant phase-sensitive test uses $J(t) = 0.08\cos(0.9t)$ to $t = 80$.

For Theorem 3, W, α, β, and Z are integrated together with DOP853. For two modes the relative and absolute tolerances are 10^{-9} and 10^{-11}. For one mode the relative tolerance is 10^{-11}. The running maximum of $\| Z \|$ uses about 3000 time points for two modes and 2×10^5 for one mode. See the two-mode script (code/phase_mm.py).

Homogeneous expanding background

To check Eq. (27), we integrate from conformal time $-(40/\rho + 20)$ to $40/\rho + 20$. Agreement is better than 10^{-8} relative for $\rho = 2{,}10$. For $\rho = 0.5$ and $k \geq 1$, the particle number falls below 10^{-7} and the comparison is limited by double precision.

For Section 4.4, phase_frw.py (code/phase_frw.py) integrates the Bogoliubov equations and phase integral in $s = \rho t$ using DOP853 with relative tolerance 10^{-10} and absolute tolerance 10^{-13}. It separates the constant-frequency carrier from the phase correction. It samples Φ on 20,001 equally spaced points in $-18 \leq s \leq 18$. The integrated coupling of the omitted tails is at most $(1 - \tanh 18)/[2(k^2 + 1)] < 2.4 \times 10^{-16}$.

The script reports peak and final occupations at $k = 0{,}1{,}3$. It uses trapezoidal quadrature on 21 linear momenta in $0 \leq k \leq 5$ and 40 logarithmic momenta in $5.3 \leq k \leq 300$. The estimated density ceilings for $\rho = 0.5{,}2{,}10$ are 1.71508×10^{-5}, 6.38252×10^{-4}, and 4.59910×10^{-3}. Refinement changes these estimates by at most 0.061%. Run python phase_frw.py --refined from code/ for 41 linear and 80 logarithmic momenta, 40,001 time samples, relative tolerance 10^{-11}, and absolute tolerance 10^{-14}. See the standard (validation/flrw_standard.txt) and refined (validation/flrw_refined.txt) outputs.

For $k \geq K = 300$, Corollary 1 gives $\Phi(k) \leq \rho/(4k^3)$ and $\Theta(k) = 1/4 \log[(k^2 + 3)/(k^2 + 1)] \leq 1/(2k^2)$. Equation (20) bounds the density tail by $\rho^2 C(K)/(96\pi^2 K^3)$, where

$$C(K) = \left(1 - \frac{\rho}{4K^3}\right)^{-2} \exp\left[\frac{\rho/(4K^3)}{K^2(1-\rho/(4K^3))}\right]. \tag{S5}$$

The tail is below 4×10^{-9} at all three rates. The sampled maxima and quadrature on $0 \leq k \leq K$ give numerical estimates of the analytic ceiling.

Moving defect and random checks

The ring calculation (code/ring.py) integrates the mode matrix with DOP853 at relative tolerance 10^{-9} and absolute tolerance 10^{-11}. It evaluates B from $\dot{V}$ in the instantaneous eigenbasis on 100 points per unit time. The phase-sensitive ring calculation (code/ring2.py) integrates W, Z, α, and β with fourth-order Runge–Kutta at step 0.004. The final occupations agree to four digits. The defect starts at $x_a = M/4$ and remains at rest for two time units after moving. The single-mode comparison is $\sum_k \sinh^2\left(1/2 \int |\dot{\omega}_k/\omega_k| \, dt\right)$ with the instantaneous frequencies ordered by size. See the Figure 3 data (code/ringscan.json).

- **Theorem 1.** The 94 accepted protocols cover one to three modes. They use $h = h_0 + a(t)P$, $h_0 = QQ^{\mathsf{T}} + cI$ with Gaussian Q and $c = 1.5$ or 2, and random symmetric P with x–p blocks. The periodic frequencies span 0.3–3 and the tanh widths 0.005–1, with h kept positive definite. Both inequalities in Eq. (15) hold to integration accuracy. The excitation-to-length-bound ratio reaches 0.9996 for the fastest quenches.
- **Theorem 2.** Twelve periodic or tanh protocols cover two and three modes with x–p blocks. The frame is $S_t = (2\gamma_t)^{1/2}$ with $2\gamma_t = h^{-1/2}|h^{1/2} J h^{1/2}|h^{-1/2}$. The pair term comes from the part of $-S_t^{-1}\dot{S}_t$ that anticommutes with J. The Heisenberg propagator uses relative tolerance 10^{-10}. The

identity $\| B \|_F = \sqrt{2}\, v_{\mathrm{FS}}$ holds to 10^{-11}. The relative excess over Eqs. (16)–(17) is at most 2×10^{-4}, consistent with the excess over Eq. (15) from time discretization near $t = 0$.

- **Theorem 3.** Forty two- and three-mode protocols use $V = V_0 + a(t)P$ with $V_0 = QQ^{\mathsf{T}} + I$ and $\| P \|_{\mathrm{op}} = 1$. The amplitude is 0.05–0.6 times the smallest eigenvalue of V_0. Periodic frequencies are 0.2–4 and tanh widths 0.03–3. The final time is 12.

The random protocols use NumPy's default_rng with seeds 7 and 11 for Theorem 1, 3 for Theorem 2, and 5 for Theorem 3. See the code index (code/README.md). For Corollary 2 we compute X in the eigenbasis of A and use central differences at step 10^{-5} for its time derivative.

S4. Coupled scalar field on a three-torus

The torus calculation (torus/coupled_torus.py) uses a real scalar field on a torus of side 2π, reference mass 1, final time 1, and spherical Fourier cutoff Λ. The addition to the squared mass is

$$U(t,x) = g(t)\left(1 + \frac{\cos x_1}{2}\right), \qquad g(t) = \frac{1-\cos(\pi t)}{2}. \tag{S6}$$

At fixed transverse labels k_2, k_3, $V = \mathrm{diag}(1 + |k|^2) + g(t)P$. The matrix P has diagonal entries 1 and entries $1/4$ between adjacent retained longitudinal labels. Blocks are grouped by $k_2^2 + k_3^2$ and all signed transverse labels count towards the real field's mode space without an extra factor of two.

In each block $Q' = P_m$ and $P_m{}' = -VQ$, with $Q(0) = F_0^{-1}$, $P_m(0) = -iF_0$, and $F_t = V(t)^{1/4}$. The block particle number in the final instantaneous vacuum convention is

$$\| \frac{F_1 Q(1) - iF_1^{-1} P_m(1)}{2} \|_F^2. \tag{S7}$$

Summing with the transverse multiplicities gives N_Λ. DOP853 uses relative and absolute tolerances 10^{-10} and 10^{-11}. The tighter check at $\Lambda = 18$ uses 10^{-12} and 10^{-13}.

Put $a_k = (1 + |k|^2)^{-1}$. The integrated drive norm is $3/2$, so Eq. (28) at finite cutoff gives the supremum-based bound $\sinh^2(3/8) \sum_{|k| \le \Lambda} a_k^2$. The pair-block estimate in Eq. (30) and the operator bound in Eq. (31) give the profile-dependent bound $W_\Lambda \sinh^2(3/8)/(3/2)^2$, where

$$W_\Lambda = \sum_{|k| \le \Lambda} a_k^2 + \frac{1}{8} \sum_{|k|,\, |k+e_1| \le \Lambda} a_k a_{k+e_1}. \tag{S8}$$

Here $e_1 = (1,0,0)$. This is the squared Hilbert–Schmidt norm of $V_0^{-1/2} P V_0^{-1/2}$ on the retained space. Both sums use the same modes as the dynamics. Cauchy–Schwarz bounds the shifted overlap by the unshifted trace. The squared spatial RMS amplitude is $9/8$, so the intermediate RMS bound is half the supremum-based bound. At $\Lambda = 18$ the bounds are 1.370508434536, 0.685254217268, and 0.678661547642, respectively. The reduction from the RMS bound to the profile-dependent bound is 0.9621%.

From the repository root, run:

```
cd torus
OPENBLAS_NUM_THREADS=1 python coupled_torus.py > coupled_torus_rerun.csv
OPENBLAS_NUM_THREADS=1 python coupled_torus.py --cutoff 18 --rtol 1e-12 --atol 1e-13 >
coupled_torus_tight_rerun.csv
python plot_field_bounds.py
```

The standard (torus/coupled_torus.csv) and tighter (torus/coupled_torus_tight.csv) CSVs record finite-cutoff particle numbers and upper bounds. The tolerance comparison checks numerical stability. Theorems 4–5 establish convergence as the cutoff is removed.